Mogomane Digashu_26304826_SS3_Part A: Report

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GitHub link: https://github.com/SuomaNuunyango/eCommerce-and-logisitcs-capstone-project/tree/main

Criteria	comment	Overall quality
Problem statement	The Problem Statement is reasonable, relevant, well-motivated and clearly describes the increasing number of returns to Stadioalot and the associated financial and operational implications, as well as proposing a data driven approach to identify potential high-risk returns. The student has identified appropriate data sources for this issue and described some practical ways in which better decisions can be made.	Good (1.5)
Motivation	The motivation for the study is clearly stated and backed up with appropriate background information in relation to Stadioalot's operations. The student also explained why reducing return rates is important for Stadioalot and have identified the financial, operational, and customer experience consequences. The proposed utilization of history to proactively inform decisions on potential returns is applicable to the business problems described above. The student could enhance this section of the report by providing verifiable evidence, or confirmation from credible business sources, as to whether your reported return	Good (2)

	rates have changed over time and if so, what has caused these changes; also, be sure to separate all returns from those which may potentially be avoided.	
Model Choices	The students' chosen two valid models for a binary classification were the Logistic Regression and the Random Forest. The comparisons are a demonstration of the student's clear depth and insight based on using the right metrics to evaluate their models; interpretating how well each model performed in terms of true positives/false positives and true negatives/false negatives. The students also acknowledged the same limitation of having comparable data splits. In this case, the Random Forest model outperformed the Logistic Regression Model by achieving an accuracy of 62.90% vs. 57.80%. Additionally, the Random Forest model had a significantly higher precision (.5973) and recall (.5838) than the logistic regression model, along with a slightly higher f1 score (.5905); test roc auc (.6844) and cross validation roc auc (.6649) than logistic regression. However stronger justification for the initial model choices and verification of identical test sets would further strengthen analysis	Good (2)
Statistical analysis	The student has a good idea of what is involved in statistical analysis when they	Good (1)

	applied techniques including stratified train/test split, cross-validation, and hyperparameter search. They also use appropriate metrics for evaluation as part of classification problem. While the random forest model appears to have performed slightly better than logistic regression based on the reported values, the inconsistency in reported false positive/false negative values creates doubt as to whether the comparison was accurate or reliable. It would be helpful if they could confirm that the confusion matrix values were consistent across both models, ensure that both models used the same test set and provide additional explanation/interpretation of the statistics generated in their study.	